Journal of Catalysis 417 (2023) 213–225

Contents lists available at ScienceDirect

Journal of Catalysis

journal homepage: www.elsevier.com/locate/jcat

Revisiting the syngas conversion to olefins over Fe-Mn bimetallic catalysts: Insights from the proximity effects

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ARTICLE INFO

Article history:
Received 14 October 2022
Revised 28 November 2022
Accepted 2 December 2022
Available online 9 December 2022

Keywords:
Bimetallic catalyst
Self-combustion method
Fischer-Tropsch synthesis
Olefins
Structure transformation

ABSTRACT

Fe-Mn bimetallic catalysts have shown great promises in direct conversion of syngas to olefins via Fischer-Tropsch Synthesis (FTS). However, due to the structural complexity of Fe-based catalysts and the interference of secondary promoters, the promotional effects of Mn on phase transition of Fe-based FTS catalysts and catalytic performance are still under debate. Herein, Fe-Mn bimetallic catalysts were prepared from two different methods, namely solution combustion method (SCM) and co-precipitation (CP) method. Fe4Mn1-SCM exhibited an excellent C_{5+} selectivity (45 %) and stability over 100 h. In contrast, the product distribution given by Fe4Mn1-CP was dominated by C_1-C_4 hydrocarbons (85 %). The superior chain growth ability of the Fe4Mn1-SCM was attributed to the abundant interfacial Fe-MnO_x structure, which retarded the oxygen removal and carbon diffusion process, resulting in less χ-Fe_5C_2 phase but more ε-$Fe_{2.2}C$ phase that was highly active for C–C coupling. Moreover, Mn-induced promotional effects were analyzed via kinetic measurements, physiochemical characterizations, *Operando* Raman-mass spectroscopy and theoretical calculations. It was found that the Fe-Mn proximity played a key role in controlling the interaction of structural and electronic impacts between Mn and Fe species. These findings may provide new insights for fabrication of Fe-based FTS catalysts with tunable product distribution.

1. Introduction

Olefins are critical components of the chemical industry, with an ever-increasing demand. Steam cracking of fractions of crude oil is the state-of-the-art technology for olefin production. The depletion of crude oil reserves, as well as the need for the energy sector to reduce its carbon footprint, has fueled global interest in seeking for chemical manufacturing processes that use renewable feedstocks. Fischer-Tropsch (FT) synthesis is one of the most efficient and sustainable technologies that enables the production of valuable chemical products from alternative carbon sources, such as biomass, coal, and solid waste, via syngas (a mixture of H_2 and CO) intermediate. The synthetic chemicals and fuels produced from FT synthesis are considered to be ultra clean in comparison to those produced from petroleum in that the former ones are free of sulfur and nitrogen-containing components, which are subsequently released into the atmosphere as SO_x and NO_x during combustion [1]. Compared with various Group VIII transition metals, the low cost of Fe makes them more economically viable for large-scale application. Furthermore, Fe catalysts exhibits an excellent water–gas shift (WGS) activity, allowing them to operate at lower H_2/CO ratios. These advantages have made Fe-based FT synthesis a compelling technology for sustainable production of olefins.

Under the typical Fe-based FT synthesis conditions, multiple iron carbides (ε-Fe_2C, Fe_7C_3, χ-Fe_5C_2, and θ-Fe_3C) can be observed, each with a different intrinsic activity and selectivity [2–4]. The nature of the formed iron carbide on the surface or the bulk of the catalyst is susceptible to environmental temperature and carbon potential (μ_C). Moreover, the transformation of various carbide phases under operational conditions presents challenges in gaining a thorough understanding of the reaction mechanisms. Given the highly structure-sensitive nature of Fe catalyst and variability of carbide phases under FT conditions, attempts have been made to tune the composition and intrinsic structure of iron carbides through doping of metal promoters, interfacing with a secondary metal (e.g., formation of alloy or mixed metal oxide), and carefully controlling pretreatment and carburization procedures [5–8]. To achieve high selectivity to olefins, Fe catalysts are usually promoted with several transition metals, such as Cu, Zn and Mn [9–11]. These promoters impose electronic and/or structural modification to the iron carbides, resulting in an increase in the intrinsic

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https://doi.org/10.1016/j.jcat.2022.12.003
0021-9517/

rate of elementary steps, improved active phase dispersion, and increased stability [12]. Transition metals with exceptional reducing abilities, such as Cu, Ru and Co can facilitate the reduction of iron oxides due to hydrogen spillover effect [13,14]. Moreover, modifications of Fe catalysts with Zn and Mn are well documented in the literature. Fe can form a homogeneous bimetallic system with Zn and Mn as MFe_2O_4 (M = Zn, Mn) spinel or $MnFeO_x$ solid solution. These mixed oxides are catalytically inactive in FT reaction but inhibit the sintering of iron particles, therefore enhancing the catalytic stability [15].

Our previous work revealed that Mn increased the dispersion of FeO_x catalyst precursor and the surface basicity of the catalyst, resulting in a boost in activity and selectivity towards olefins. The strong Fe-Mn interaction, on the other hand, imposed a dramatic effect on stabilizing the $Fe_{1-x}Mn_xO$ phase and inhibiting the formation of active Fe_xC [16,17]. A Fe/Mn molar ratio of 4 is found to grant the maximum yield of olefin. Recently, Ding et al [6]reported that a hydrophobic core–shell FeMn@Si catalyst exhibited a total olefin selectivity of 83.3 % with an extremely low selectivity to C1-byproducts (CO_2 and CH_4). Mn was found to suppress the oxidation of iron carbides from in situ generated water during reaction. According to the literature survey, Mn is widely regarded as an electronic and structural promoter. However, the interference of other factors, such as secondary promoters (e.g., Na and K) and catalyst preparation methods may lead to contradictory conclusions on the promotional effects of Mn on FTS activity and product selectivity.

The proximity effect is the tendency of catalytic performance to be affected by the distance between two active sites in a bifunctional catalyst. The proximity between active sites of a bifunctional Co-Fe_3O_4 CO_2 hydrogenation catalyst, for instance, has been shown to be extremely important. The catalyst system contains of active sites for the reverse water gas shift reaction (Fe_3O_4), CO dissociation and C–C coupling (Co^0 and χ-Fe_5C_2) [18]. Owing to their close proximity, CO spillover from Fe_3O_4 sites to CO^0 sites is favored, resulting in a high CO coverage on Co^0 sites and increased C_{2+} hydrocarbons yield. Unlike bifunctional catalyst systems, the Mn promoters are inactive for FTS reaction but have an indirect effect on the catalytic process by tuning the geometric and electronic properties of iron phase. As a result, the proximity between Mn promoter and Fe is expected to have a significant impact on the phase composition of Fe and the FTS performance. To the authors' best knowledge, studies of proximity effect on Fe-Mn bimetallic FTS catalysts were rarely reported in literature.

The method of catalyst preparation is critical in tuning important catalyst properties, such as particle size, surface area, and pore structures. Fe-based FT catalysts are typically prepared using wet synthesis methods, such as co-precipitation, hydrothermal, and impregnation, which are then dried and calcined. The quality of the prepared catalyst is sensitive to numerous parameters in each individual step, resulting in a poor reproducibility. Moreover, the nucleation and growth rate of the primary crystals are sensitive to a variety of factors (e.g., temperature, pH, stirring rate, composition of the precursor solution), which further brings challenge to obtain highly crystalline and homogeneous products.

In comparison to these traditional methods, solution combustion synthesis (SCS) is a facile and cost-efficient method to synthesize nanomaterials with homogeneous composition, close proximity of different components, and controlled properties for a wide range of applications, including heterogeneous catalysis, photocatalysis, energy storage, and so forth [19,20]. In the SCS process, a thermally initiated and self-propagated combustion takes place within a redox mixture of an organic fuel, also known as the reducing agents (e.g., citric acid, urea, glycine) and an oxidant (usually metal nitrate) based on the fact that the energy released from the exothermic decomposition of the redox mixture can sustain the combustion synthesis [21]. As a result, the SCS process can be carried out with less energy input than the high-temperature calcination step. In addition, the gases (e.g., H_2, CH_4, CO) released during the thermal decomposition of fuel can reduce the metal precursors and favor the formation of porous products with a high surface area, which is conducive to heat and mass transfer in catalytic reactions. Studies adopting SCS method for preparing catalysts for methanol synthesis, dry reforming of methane, FT reaction and CO_2 hydrogenation have been reported in literature [22–25].

In this study, Fe-Mn bimetallic bulk catalysts were prepared by SCS method. The as-prepared catalysts were used for directly converting CO to olefins via FT reaction and compared with the ones prepared using the traditional co-precipitation method. It was discovered that the Fe-Mn bimetallic catalyst prepared by SCS method improved the nanoscale proximity between Fe and Mn, resulting in a product distribution shift towards higher carbon chains with a high olefin-to-paraffin ratio.

2. Materials and methods

2.1. Catalyst preparation

Fe-based catalysts were prepared by solution combustion synthesis method. As for Fe-Mn bimetallic catalyst, Fe and Mn precursor salts ($Fe(NO_3)_3·9H_2O$ and 50 wt% $Mn(NO_3)_2$) were dissolved in deionized water at a Fe:Mn molar ratio of 4:1 to form a metal precursor solution. Citric acid was used as the reductant, and was added drop wisely to the metal precursor solution at a molar ratio of citric acid:(Fe + Mn) = 2:1. The resulting mixture was stirred under water bath at 50 °C for 2 h to form a homogeneous and transparent gel. The resultant gel was then transferred to a muffle furnace setting at 350 °C to initiate the combustion and calcined for 4 h to yield a loose and fluffy catalyst. For comparison, Fe-based catalysts were also prepared using a co-precipitation method. Here, 0.01 mol of $Fe(NO_3)_3·9H_2O$ and 0.0025 mol of 50 wt% $Mn(NO_3)_2$ were dissolved in 50 mL ethylene glycol, and 0.025 mol of Na_2CO_3 was dissolved in 30 mL deionized water, then the Na_2CO_3 solution was added drop wisely to the metal precursor solution under −10 °C. The slurry was aged for 1 h and washed by 500 mL deionized water for 5 times before filtration. Finally, the filter cake was dried at 60 °C overnight and calcined at 450 °C for 4 h. In this study, the Fe-Mn bimetallic catalysts prepared using solution combustion method and co-precipitation method are denoted as Fe4Mn1-SCM and Fe4Mn1-CP, respectively.

The activated and spent samples for further characterization were prepared by a certain procedure. 100 mg of the fresh samples were activated in syngas (5 %CO/5 %H_2/N_2, 25 mL/min) at 350 ℃ for 5 h. The reaction was conducted in syngas (45 %CO/45 %H_2/N_2, 25 mL/min) at 280 °C after activation, and the spent catalysts were collected after 10 h reaction.

2.2. Catalyst characterization

The metal content of different samples was tested by inductively coupled plasma optical emission spectrometry (ICP-OES, Agilent 5110). X-ray diffraction (XRD) patterns were collected using a PANalytical X'pert-3 Powder diffractometer operating with Cu Kα radiation at 40 kV and 40 mA. The patterns were captured using a step of 0.02° and a scanning rate of 10°/min within a 2θ range of 10°-80°. Pore structure of the catalysts were measured by N_2 physisorption at −196 °C using a Micromeritics ASAP-2460 adsorption instrument. The specific surface area (S_{BET}) was calculated using the Brunauer-Emmett-Teller (BET) method. Surface morphology and surface elemental distribution were characterized using a scanning electronic microscope (SEM, Gemini 300, ZEISS) paired with an energy-diffusive X-ray spectrometer. The residual carbon content of the as-prepared catalysts was determined using an elemental analyzer (VARIO EL III). High-resolution Transmission Electron Microscope (HRTEM) experiments were carried out using an FEI TalosF200x electron microscope equipped with a Super-X EDS spectrometer. The ^{57}Fe Mössbauer spectra (MES) of the activated and spent catalysts were recorded at 298 K on a Wissel 1550 electron mechanical spectrometer (Wissenschaftliche Elektronik GmbH) equipped with a ^{57}Co/Pd irradiation source.

The reduction behavior of the as-prepared catalysts was analyzed by H_2 temperature programmed reduction (H_2-TPR) on a micro fixed-bed reactor coupled with a thermal conductivity detector (TCD). Approximately 50 mg of catalyst samples were pretreated in pure Ar at 150 °C for 2 h to remove the physical adsorbed moisture. The H_2-TPR experiments were conducted under 10 % H_2/Ar flow at 25 mL/min at temperatures ranging from 100 to 800 °C with a ramping rate of 10 °C/min. The surface carbonaceous species was by temperature programmed hydrogenation (TPH) on the same micro fixed-bed reactor as in H_2-TPR with a mass spectrometer (HPR-20, Hiden Analytical Ltd.) for detecting CH_4 signal (m/z = 16). 20 mg of spent catalyst samples were degassed under pure Ar at 150 °C for 1 h, after cooling down to room temperature, the catalysts were switched into 10 % H_2/Ar at 25 mL/min in the temperature range of 100–900 °C with a ramping rate of 10 °C/min. Structural and compositional features of the as-prepared catalysts were characterized using a Raman spectroscopy (LabRAM HR, Horiba J.Y.) with a visible 514.5 nm Ar + laser citation. The spectra were calibrated using the characteristic of single crystal silicon at 520.7 cm^{-1}.

To study the structural evolution at catalytically relevant conditions, the as-prepared catalysts were characterized using the same Raman spectroscopy coupled with a mass spectrometry. The catalysts were activated at a rate of 5 °C/min under syngas (5 %CO/5 % H_2/Ar) from 30 °C to 350 °C and maintained for 5 h. Subsequently, the catalysts were cooled to 100 °C and pressurized to 2.0 MPa using the same syngas. Afterwards, CO hydrogenation reaction was taken place at 300 °C and lasted for 10 h. Several typical products such as CO_2 (m/z = 44), CH_4 (m/z = 15), C_2H_4 (m/z = 27) and C_3H_6 (m/z = 41) evolved during reaction were simultaneously detected.

2.3. Catalytic performance evaluation

The FT reaction of the Fe-based catalysts were tested in a fixed-bed reactor system, as previously described [16]. Catalyst samples (25 mg) diluted with SiC (75 mg) were loaded in the reactor and activated at 350 °C for 5 h in a syngas (5 %CO/5 %H_2/N_2) flow and then tested under 280 °C, 2.0 MPa in a syngas (45 %CO/45 %H_2/N_2) flow, and gas hourly space velocity (GHSV) = 7500 h^{-1}. Gaseous products (H_2, N_2, CO, CO_2, C_1-C_7 hydrocarbons) were analyzed using an online GC (Shanghai Ruimin, GC2060) equipped with a TCD and two FIDs. Liquid products (C_5-C_{20} hydrocarbons) collected from a cold trap were analyzed using a gas chromatography-mass spectroscopy (GC/MS, GC: 7890A, MS: 5975C, Agilent Technologies) equipped with a HP-5 capillary column. CO conversion, CO_2 selectivity, hydrocarbon selectivity and space time yield (STY) of hydrocarbons were calculated according to Eq. (1) ~ Eq. (4), respectively.

$$\mathrm{CO\ \ conversion} = \frac{\mathrm{CO_{inlet}} - \mathrm{CO_{outlet}}}{\mathrm{CO_{inlet}}} \times 100\% \tag{1}$$

$$\mathrm{CO_2\ \ selectivity} = \frac{\mathrm{CO_{2outlet}}}{\mathrm{CO_{2outlet}} + \sum \mathrm{m} \times \mathrm{C_mH_n}} \times 100\% \tag{2}$$

$$\mathrm{Hydrocarbon\ \ selectivity(C-mol.\%)} = \frac{\mathrm{m} \times \mathrm{C_mH_n}}{\sum \mathrm{m} \times \mathrm{C_mH_n}} \times 100\% \tag{3}$$

$$STYg \cdot kg^{-1} \cdot h^{-1} = \frac{CO_{inlet} \times V_{CO} \times COconversion}{V_m \times m_{cat}} \times C_mH_nselectivity \times M_{C_mH_n} \tag{4}$$

Where V_m and m_{cat} are the molar volume of an ideal gas at standard condition (V_m = 22,400 mL/mol) and mass of the catalyst, respectively. $M_{C_mH_n}$ is the molecular weight of component C_mH_n.

3. Results and discussion

3.1. Textural properties of the as-prepared catalysts

The textural properties and surface morphology of the as-prepared catalysts were examined using N_2 physisorption and SEM imaging. Two calcined catalysts exhibit distinct N_2 adsorption–desorption isotherms (Fig. S1). Fe4Mn1-SCM shows a steep uptake at higher relative pressure (p/p^0 > 0.6) with a final saturation plateau and a large hysteresis loop, which are typical characteristics of mesoporous material corresponding to a Type IV(a) isotherm according to the IUPAC classification. In comparison, Fe4Mn1-CP shows a much smaller uptake in the whole pressure range with a weak hysteresis loop above p/p^0 = 0.6, corresponding to a type III isotherm which is commonly seen in the nonporous or microporous materials. The pore size distribution (PSD) shows that most of the pores of Fe4Mn1-SCM fall into the size range of 7–11 nm, whereas Fe4Mn1-CP does not give a recognizable peak on the PSD curve, indicating a nonporous structure. The total pore volume, average pore size and specific surface area calculated according to the Barrett-Joyner-Halenda (BJH) method are summarized in Table 1. As expected, the catalyst prepared using solution combustion method has a higher specific surface area and pore volume than those prepared using co-precipitation method. The release of a substantial amount of gaseous products during combustion enables the development of porosity and increases the total surface area [20]. In addition, the particle size of the two catalysts also showed a clear difference. Such difference in particle size may result in distinct nano proximity between the active metal and promoter or support, which further affects the electronic and geometric properties of the catalysts [26–29]. In this case, smaller nanoparticle size of Fe4Mn1-SCM increases the relative surface area, and may shorten the interparticle distance, causing more geometric and electronic contact between Fe and Mn species.

The microstructure of two as-prepared catalysts revealed that catalyst prepared via combustion method exhibited a porous, foamy morphology, whereas the catalyst prepared by co-precipitation method had a compact aggregation of fine nanoparticles (Fig. 1). Fe and Mn elements are evenly distributed in both two samples without noticeable agglomeration (Fig. S2). The carbon residue of the as-prepared catalyst prepared via combustion method was determined using an elemental analyzer. As shown in Table S1, only 0.11 wt% of carbon remained in the as-prepared catalyst, manifesting a complete combustion of citric acid.

Phase compositions and crystal structures of the as-prepared Fe4Mn1 catalysts were examined using XRD and Raman spectroscopy. As depicted in Fig. 2, Fe4Mn1-CP exhibits pure α-Fe_2O_3 phase with diffraction peaks at 2θ = 24.1°, 33.4°, 35.6°, and 49.6°, corresponding to (012), (104), (110), and (113) planes of α-Fe_2O_3 (PDF#84-0308). In comparison, Fe4Mn1-SCM shows diffraction peaks at 2θ = 30.2°, 35.6°, 43.2°, 57.2°, and 62.8°, matching diffraction patterns of both γ-Fe_2O_3 (PDF#39-1346) and Fe_3O_4 (PDF#65-3107). It should be noted that no features related to Mn

Table 1
Textural properties of the as-prepared Fe4Mn1 catalysts.

	BET surface area (m^2/g)	Pore volume (cm^3/g)	Average pore size (nm)	Average grain size[a] (nm)	Metal content (wt%)[b]		
					Fe	Mn	Na
Fe4Mn1-SCM	65	0.197	8.6	10.3	55.269	9.893	–
Fe4Mn1-CP	48	0.109	8.5	18.9	53.698	9.574	0.004

[a] The grain size is calculated using the Scherrer equation.
[b] Metal content from ICP-OES.

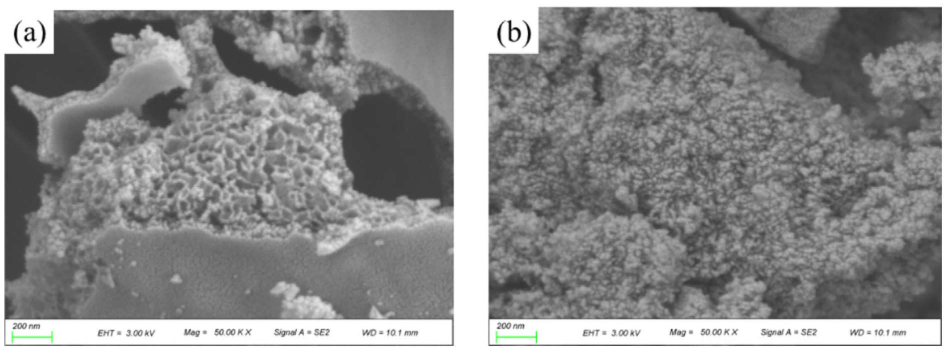


Fig. 1. Surface morphology of the as-prepared Fe4Mn1-SCM (a) Fe4Mn1-CP (b).

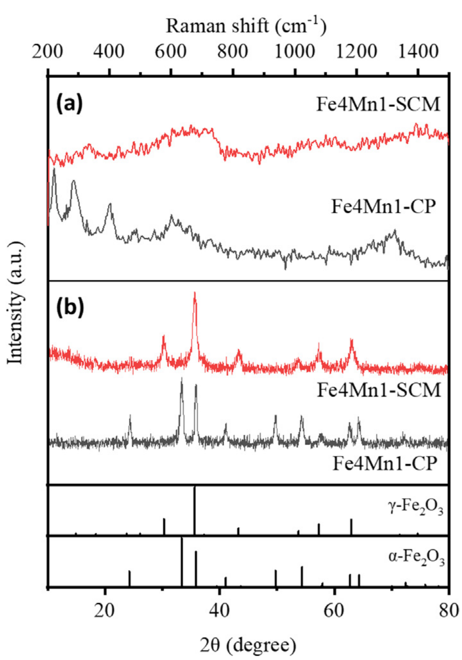


Fig. 2. Raman spectra (a) and XRD patterns (b) of Fe4Mn1-SCM and Fe4Mn1-CP.

were identified in the diffraction patterns, suggesting that MnO_x were presented as amorphous form. The average grain size calculated from Scherrer's equation confirmed that SCM rendered a more dispersed Fe-Mn bimetallic precursor. Since γ-Fe_2O_3 shares the same diffraction patterns with Fe_3O_4, the phase compositions of the as-prepared catalysts were further analyzed using 514 nm Raman. For Fe4Mn1-CP, strong bands were observed at 221 cm^{-1}(A_{1g}), 285 cm^{-1}(E_g), 401 cm^{-1}(E_g), and 1318 cm^{-1}(2LO), which were indicative of α-Fe_2O_3. In the case of Fe4Mn1-SCM, an intense peak at 331 cm^{-1} and a broad band around 700 cm^{-1} associated with T_{1g} and A_{1g} vibration modes of γ-Fe_2O_3 were observed [30,31]. From the characterization results of the as-prepared Fe4Mn1 catalysts, it can be inferred that solution combustion method produces a γ-Fe_2O_3 phase with loose and porous structure, resulting in close proximity between Fe and Mn.

3.2. Phase composition of the reduced and carburized catalysts

The reducibility of the as-prepared Fe4Mn1 catalysts prepared via two different methods were characterized using H_2-TPR. As shown in Fig. 3, both catalysts display different stages of phase transition during the reduction process. The as-prepared Fe4Mn1-CP shows three major H_2 consumption peaks, with the first peak (near 280 °C) attributed to the reduction of defective $MnO_2 \rightarrow Mn_2O_3$; the second peak (near 355 °C) corresponding to the overlapped reduction events of α-$Fe_2O_3 \rightarrow Fe_3O_4$ and $Mn_2O_3 \rightarrow Mn_3O_4$; and the broad reduction peak in the range of 400–700 °C corresponding to the consecutive reduction of $Fe_3O_4 \rightarrow FeO \rightarrow Fe^0$ [16,32]. For Fe4Mn1-SCM, the reduction step of $MnO_2 \rightarrow Mn_2O_3$ is

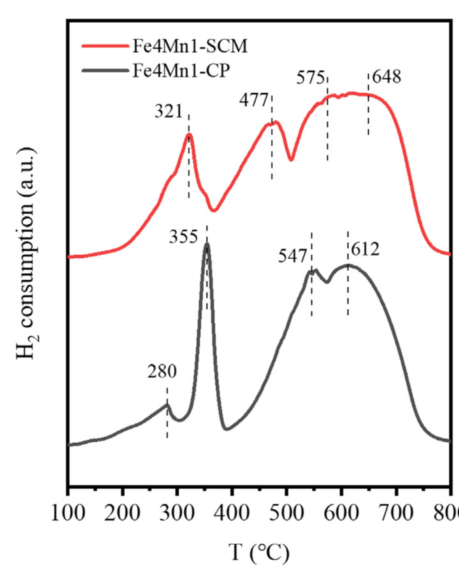


Fig. 3. H_2-TPR profile of the as-prepared Fe4Mn1 catalysts.

overlapped with that of γ-Fe_2O_3 → Fe_3O_4, resulting in a reduction peak observed at 321 °C. A shoulder peak at 477 °C can be ascribed to the reduction step of Mn_3O_4 → MnO in the Fe-Mn mixed oxide [32]. Further reduction of Fe_3O_4 to Fe^0 via FeO intermediate occurs at higher temperatures (550–650 °C), suggesting that the FeO becomes more resistant to reduction when an appropriate amount of Mn is included. Through in-situ XRD, we have revealed that the formation of highly stable $Fe_xMn_{1-x}O$ phase during the reduction process of Fe-Mn mixed oxide is responsible for the retarded reduction process [16,17]. Compared to co-precipitation method, solution combustion method gives a much smaller particle size, which gives rise to a intimate bonding between Fe and Mn during preparation process and $Fe_xMn_{1-x}O$ phase during reduction process. Overall, these findings clarify that Fe4Mn1 catalyst prepared by solution combustion method shows a weakened reducibility compared to the Fe4Mn1 catalyst prepared by co-precipitation.

The bulk phases of activated catalysts were identified using MES. As demonstrated in Fig. 4 and Table S2, the spectrum of Fe4Mn1-CP is well-fitted by two doublets and three sextets. The two doublets with the IS of 0.21 and 1.11 mm/s represent the superparamagnetic Fe^{3+} and Fe^{2+} species, respectively, whereas the three sextets with the Hhf values of 210.88, 184.55 and 110 kOe are corresponded to the χ-Fe_5C_2. For Fe4Mn1-SCM, another sextet with the Hhf value of 201.17 kOe appeared, and it may be assigned to the θ-Fe_3C phase, which is stable at relatively high temperatures. The quantitative results preliminarily confirmed that χ-Fe_5C_2 is the dominant species in the activated catalysts, but Fe4Mn1-CP has a greater carbide content (92.4 % of χ-Fe_5C_2) than Fe4Mn1-SCM (62.7 % of χ-Fe_5C_2 and 4.9 % of θ-Fe_3C). A small amount of Fe_3C can be seen in activated Fe4Mn1-SCM, possibly due to higher temperatures and lower μ_C during activation which help the formation of carbon-poor carbide species more favorable [33]. Iron carbides are intermetallic compounds in which carbon atoms occupy the interstices between close-packed iron atoms [34,35]. Although several studies have shown that iron oxides, such as Fe_3O_4, may be the intermediates of Fe carbides, the lattice O atoms may cause a significant driving force in carbon diffusion [36,37]. As a result, the carburization ability always increases with the reduction degree grows. According to the H_2-TPR results, Mn stabilizes partially reduced Fe oxides, especially $Fe_xMn_{1-x}O$, therefore the decreased carbide concentration in Fe4Mn1-SCM can be attributed to the insufficient reduction during the activation process.

Composition and phase structure of the spent catalysts were analyzed using XRD and TEM. As shown in Fig. 5, the sharp diffraction peak at 26.5° is attributed to the quartz wool mixed with the spent catalysts. The diffraction patterns of two spent catalysts differ significantly. Fe4Mn1-CP shows exclusively features of χ-Fe_5C_2 phase, but the diffraction patterns from the spent Fe4Mn1-SCM catalyst are more complex, containing mixtures of χ-Fe_5C_2, $Fe_xMn_{3-x}O_4$, and $Fe_xMn_{1-x}O$ phases. The presence of χ-Fe_5C_2 phase in both catalysts can be further confirmed by comparing the FFT images to the XRD patterns, as shown in Fig. 6 and Table S3. Meanwhile, the $Fe_xMn_{3-x}O_4$ phase is also identified in the TEM images of Fe4Mn1-SCM, and is in consistence with the XRD data. Surprisingly, isolated Mn_5O_8 phase was discovered in Fe4Mn1-CP. Xiang et al. identified an oxidic phase with with Mn^{2+} and Mn^{4+} mixed valence states over Co-based catalysts, which formed under low H_2/CO conditions (P_{H2}/P_{CO} = 1.5) from MnO_x and may further decompose to develop smaller MnO with cubic rock-salt structure [38]. Thus, it is postulated that χ-Fe_5C_2 is one of the main active phases in both catalysts and strong Fe-Mn interaction in Fe4Mn1-SCM favors the formation of mixed Fe-Mn oxides rather than isolated MnO_x. In general, χ-Fe_5C_2 is commonly recognized as the active phase for CO dissociation and C–C coupling reaction in Fischer-Tropsch reaction [39]. The Fe-Mn mixed oxides ($Fe_xMn_{3-x}O_4$, and $Fe_xMn_{1-x}O$) can impede the reduction and carburization of iron nanoparticles due to enhanced Fe-O bond strength and unfavorable Fe-C bonding [40].

The composition of spent catalysts was further analyzed using MES, and the results were shown in Fig. 7 and Table S4. From the quantitative results, the spent Fe4Mn1-CP contains 82.7 % χ-Fe_5C_2 with the remainder being superparamagnetic Fe^{3+}. Similar to the XRD findings, no iron carbide or iron oxide phases other than χ-Fe_5C_2 are detected in the MES of Fe4Mn1-CP. The spent Fe4Mn1-SCM shows the features of two iron carbides (ε-$Fe_{2.2}C$ and χ-Fe_5C_2), Fe_3O_4 and superparamagnetic Fe^{3+}/Fe^{2+}. Compared with the activated catalysts, the carbide content in both catalysts decrease, while the proportion of Fe^{3+} relative to Fe^{2+} become higher, revealing that the catalysts may suffer some oxidation throughout the reaction. The content of χ-Fe_5C_2 phase in Fe4Mn1-SCM is 52 %, which is significantly lower than that in Fe4Mn1-CP (82.7 %). Besides χ-Fe_5C_2, trace amount (5.1 %) of carbon-rich ε-$Fe_{2.2}C$ is

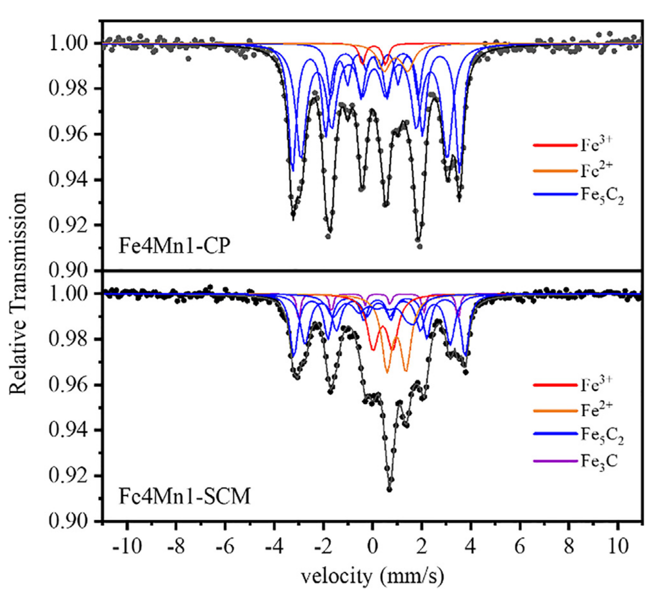


Fig. 4. MES spectra of the activated Fe4Mn1 catalysts.

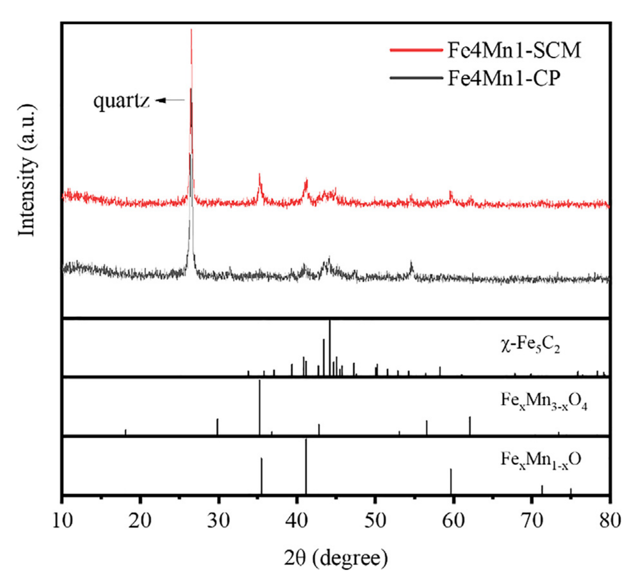


Fig. 5. XRD patterns of the spent Fe4Mn1 catalysts.

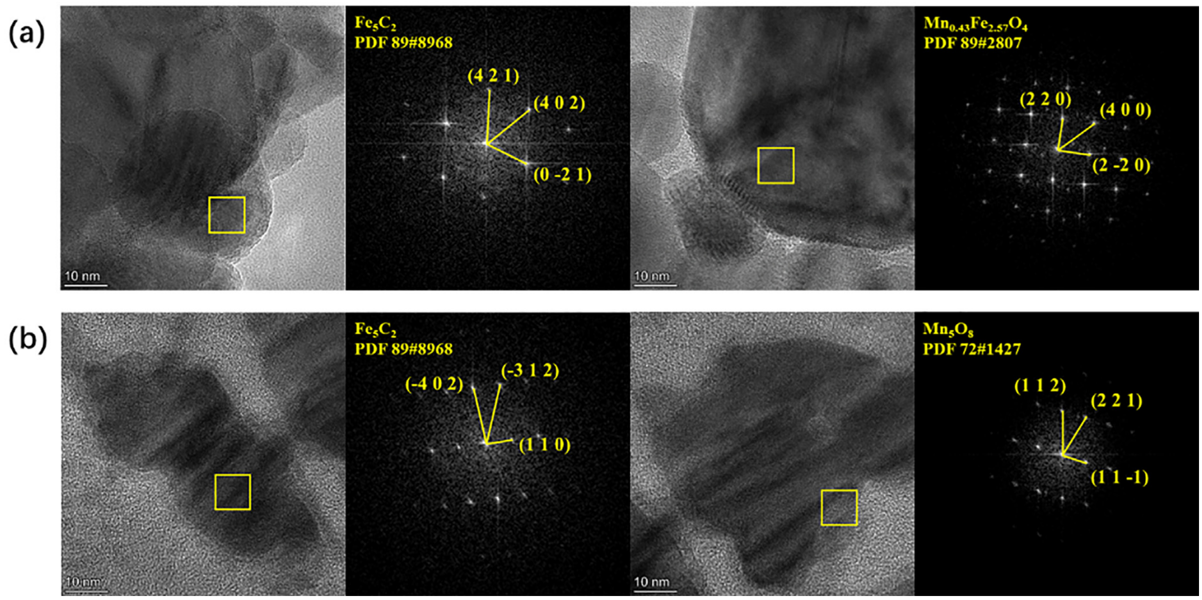


Fig. 6. High resolution TEM images and FFT analysis on (a) Fe4Mn1-SCM and (b) Fe4Mn1-CP.

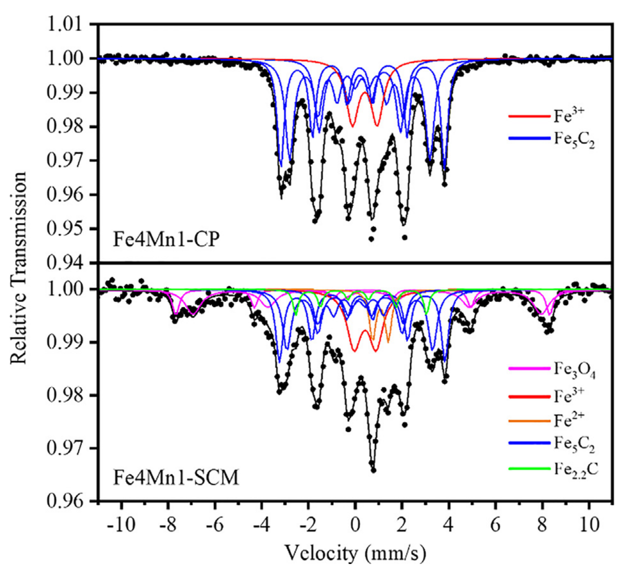


Fig. 7. MES spectra of the spent Fe4Mn1 catalysts.

detected. The latter is known as the most active phase for C–C coupling reaction, but is metastable during reaction and susceptible to transform into χ-Fe_5C_2 phase above 250 °C [2]. Mogorosi et al. [41] proposed that strong Fe-Si interaction deriving from Fe-O-Si bonding structure adjusted the balance between carbon diffusing rate and lattice expansion, and may help the transformation from FeO to Fe_2C. Our previous study also revealed the presence of χ-Fe_5C_2 and ε-$Fe_{2.2}C$ phases in the spent Fe-Mn catalysts, with the proportion of the latter increasing when more Mn was added, indicating that Mn hindered the phase transition of ε-$Fe_{2.2}C \rightarrow \chi$-$Fe_5C_2$ [16]. Interfacing iron oxide with noble metals (e.g., Pt and Au) was reported to be a promising strategy to fabricate a highly stable ε-$Fe_{2.2}C$ phase in FT reaction conditions, owing to the appropriate electronic interaction between the highly dispersed noble metal nanoparticles and the iron carbides as well as their intimate contact [42].

It is assumed that Fe oxides can also become the intermediates of Fe carbides, and the carburization process can be divided into two steps: (1) the removal of lattice O or dissociated O atoms from CO and, (2) the deposition and further diffusion of the C atoms [43]. According to the above H_2-TPR and MES results, it can be clearly found that Fe-Mn intimate contact hinders the reduction of Fe species and influences the FeC_x composition in the activated and spent catalysts. MohammadReza Elahifard et al. [44]reported an approach of tuning the neighbor environment by changing different substitutional configurations and found that the nearest neighbor of active atom can influence the adsorption behavior through a ligand effect. Based on this finding, DFT calculations were employed to verify the difference in carburization ability under different Fe-Mn interactions, based on Fe_3O_4 (311) and FeO (200) surfaces. As illustrated in Fig. 8, the doping of Mn can significantly increase the energy of oxygen vacancy formation (ΔE_{ov}) in both Fe_3O_4 and FeO. Similar findings were proposed by Han et al. [40] before, however, Liu et al. [45]found Mn may promote the reduction by the presence of oxygen vacancy in MnO. Mn can function as both an electronic additive and a structural additive, so the effects on Fe species reduction are complex. Previous studies have suggested that a small amount of Mn could promote FeO_x reduction, while excessive Mn may inhibit FeO_x reduction [46]. Hence, the controversy over the role of Mn can be attributed to different Mn amount and structural models (interfacial or doped structure) . Obviously, the carburization process on FeO surfaces is easier than Fe_3O_4 surfaces, as confirmed by the more negative values of ΔE. Furthermore, by comparing the Mn-modified Fe_3O_4 and FeO with different Fe-Mn proximity, a more uniform distribution of Mn in FeO_x may limit the formation of oxygen vacancies even further. This suggests that high Fe-Mn proximity may strengthen the Fe-O bonding and be detrimental to the reduction of Fe oxides [40].

Permeation of carbon atom into the oxygen vacancy appears to be significantly easier on the Mn-decorated surfaces, with lower energy change (ΔE_C) in this process. Unlike ΔE_{ov}, the ΔE_C did not linearly depend on the doping amount and dispersion of Mn atoms. The ΔE_C of both Fe_3O_4 and FeO are endothermic by 1.80 eV and 1.65 eV, respectively. A certain concentration of Mn obviously decreases the energy change to a negative value (-0.31 and −3.70 eV for Fe_3O_4_Mn_3O_4 and FeO_MnO, respectively). However, strong Fe-Mn interactions in $(Fe,Mn)_3O_4$ and (Fe,Mn)O cause a modest increase in ΔE_C to 0.84 and −3.51 eV, respectively. Combining these two steps, the total reaction energy (ΔE_{total}) of

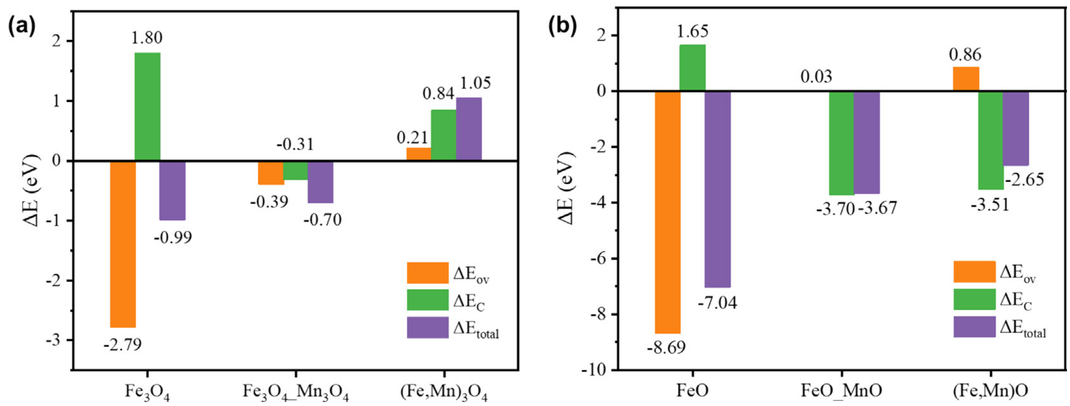


Fig. 8. Energy change during the carburization process on pure and Mn-modified (a) Fe_3O_4 (3 1 1) and (b) FeO (2 0 0).

pure and Mn-promoted FeO_x shows comparable difference. In some ways, Mn, like many alkali metal promoters, could also act as an electron donor to strengthen the Fe-C bonding and to enhance the CO adsorption and dissociation through electronic interactions, and such promotional effects may be different when the Fe-Mn distribution becomes uniform, most likely due to some geometric factors, such as the stabilization of some inert phases or the change of Fe coordination number [17]. In conclusion, the negative effect of Mn on carburization is mainly attributed to the limitation of oxygen removal ability, and excessive Fe-Mn interaction will further weaken the carbon diffusion into bulk phase, which finally determine the induced modulation of active structure.

3.3. Carbon deposition behavior

The elemental distribution of the spent catalysts is shown in Fig. 9 using energy-dispersive X-ray (EDX) mappings. The presence of Fe element can also be seen in most distribution areas of C for spent Fe4Mn1-SCM, with only a small quantity of isolated C. There are also certain regions with homogeneous distributions of Fe, Mn and O, which might be attributed to the Fe-Mn mixed oxide phases. In contrast to Fe4Mn1-SCM, a significant amount of isolated C is dispersed around Fe for spent Fe4Mn1-CP, signifying severe carbon deposition on the Fe surface. Furthermore, the distribution of Mn is not highly consistent with that of Fe, resulting in the presence of additional MnO_x phases in this catalyst. These

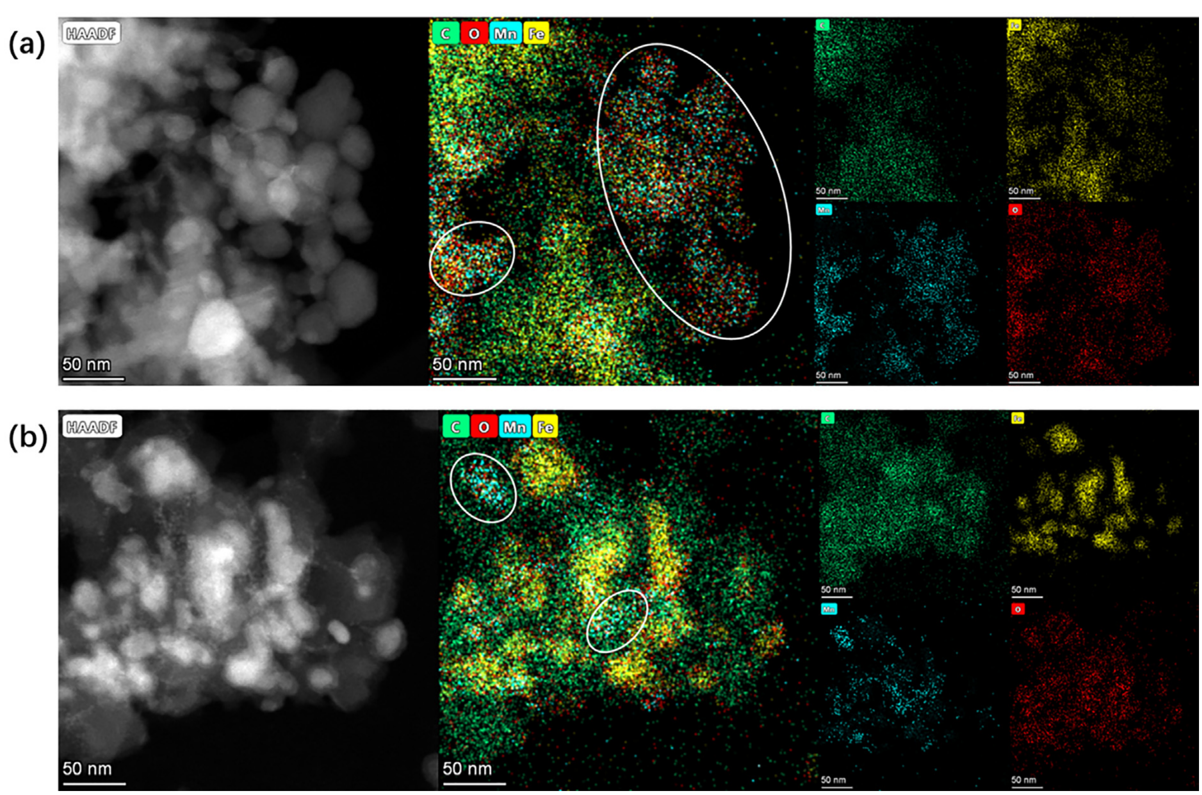


Fig. 9. EDX elemental mapping images of the spent (a) Fe4Mn1-SCM and (b) Fe4Mn1-CP; Fe (yellow), Mn (blue), C (green), O (red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

results imply that Fe4Mn1-SCM exhibits a greater Fe-Mn proximity and favors the formation of mixed oxide phases. The stabilization of such phases may hinder the carburization process but also inhibit the carbon deposition, resulting in the most of carbon existing in the form of FeC_x, whereas the co-precipitated catalyst may weaken the compounding of FeO_x and MnO_x, resulting in more carbides and carbon deposits during the reaction.

Reaction-driven phase transformation and surface reconstruction of active iron species is closely related with the interaction of surface carbons and the surface/bulk iron species. Surface carbonaceous species on Fe4Mn1 catalysts were identified with TPH-MS. Fig. 10 depicts the spectra of methane evolution rate for two spent Fe4Mn1 catalysts, as well as the associated deconvoluted spectra. The curve-fitting of the TPH profiles results in six peaks, denoted as α, β, γ_1, γ_2, δ_1, and δ_2, which can be assigned to atomic carbon (C_α, below 350 °C), amorphous surface polymethylene chains or films (C_β, 350–450 °C), iron carbides (C_γ, 500–600 °C) and graphitic carbon (C_δ, above 600 °C) [47,48]. Table 2 shows the deconvoluted peak components and their corresponding percentage compositions. Compared to Fe4Mn1-SCM, the peak maxima of peak C_β in Fe4Mn1-CP shifts to higher temperature, implying that the surface carbons are more resistant to hydrogenation. It is also evident that the peak area percentages of C_γ and C_δ in Fe4Mn1-CP are higher than those in Fe4Mn1-SCM; moreover, the similar trend is observed in the peak area ratio of δ_2/δ_1. Compositional difference of the iron carbides (ε-$Fe_{2.2}C$ and χ-Fe_5C_2) can be observed in terms of peak area of γ_1 and γ_2. As listed in Table 2, the γ_2/γ_1 ratio in Fe4Mn1-CP is higher than that in Fe4Mn1-SCM. Surface composition of the spent catalysts were further examined using XPS. Deconvolution of Fe 2p XPS survey (Fig. S4) shows substantial features of FeO_x in both spent catalysts with negligible iron carbides. This can be due to the surface oxidation during passivation pretreatment prior to sample analysis. Surface elemental analysis from XPS measurement confirms that Fe4Mn1-CP has a higher C/Fe ratio than Fe4Mn1-SCM (Table 2). These findings indicated that Fe4Mn1 catalyst prepared using solution combustion method exhibits a lower extent of carburization and less severe carbon deposition.

Operando Raman spectroscopy coupled with MS was performed to further track the transformation of iron species during activation and reaction stages. To investigate the activation process, the as-prepared Fe4Mn1 catalysts were reduced with a flow of 5 % H_2/5 %CO/90 %N_2 under atmospheric pressure from room temperature to 350 °C, and held for 5 h. Raman spectra and the MS signal of the reactants (CO and H_2) and products (CO_2, H_2O, CH_4, C_2H_4, and C_3H_6) during activation with increasing temperature and time-on-stream were illustrated in Fig. 11. In-situ Raman measurements show that the intensity of features from α-Fe_2O_3 (291 cm^{-1}, 398 cm^{-1}, 1312 cm^{-1}) and γ-Fe_2O_3 (650 cm^{-1}, 719 cm^{-1}) phases in both Fe4Mn1 catalysts decrease substantially upon exposure to reaction gas at 200 °C.

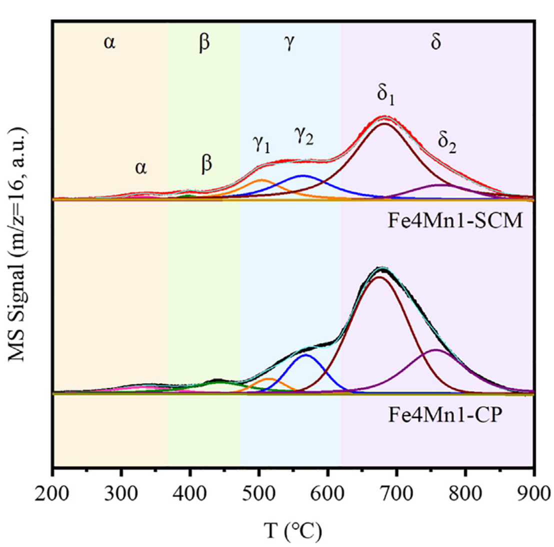


Fig. 10. TPH profile of the spent Fe4Mn1 catalysts.

Observing the accompanying fluctuations in gaseous effluents during this stage reveals a continuous consumption of CO and H_2 as well as buildup of CO_2 and H_2O for Fe4Mn1-SCM. Fe4Mn1-SCM exhibited two well-resolved CO_2 peaks centered at 288 °C and 350 °C, corresponding to the reduction of iron oxides and the subsequent carburization. In comparison, Fe4Mn1-CP showed a major CO_2 production peak centered at 328 °C, followed by a broad peak when the temperature stabilized around 350 °C. It should be noted that both CO and H_2 can reduce iron oxides, with the former being the stronger reducing agent. The earlier appearance of the first CO_2 generation peak in Fe4Mn1-SCM implies a faster initial reduction rate, which is consistent with the H_2-TPR result. Interestingly, despite higher consumption rates of H_2 and CO, the peak intensity of H_2O does not increase significantly in the case of the Fe4Mn1-CP catalyst. This is due to an increase in H_2O consumption in the water–gas-shift reaction, which corresponds to the faster CO consumption rate than H_2 observed in Fig. 11(d).

As the temperature further increased to 350 °C, a surge in peak intensities of CO_2, CH_4, C_2H_4 and C_3H_6 was observed accompanied by a rapid plummet of H_2 peak intensity. It should be noted that at the initial stage of 350 °C maintaining, the peak intensities of CH_4, C_2H_4, and C_3H_6 given by Fe4Mn1-SCM were lower than those of Fe4Mn1-CP, implying that Fe4Mn1-CP exhibited a substantially higher carburization rate of during activation and generate more active Fe carbide species. After holding temperature at 350 °C for 5 h, these signals for both catalysts show some decline, implying a decrease of the reactivity at atmospheric pressure, possibly due to the accumulation of carbon deposits. This is further corroborated by the appearance of carbonaceous species observed at 1350 and 1580 cm^{-1}. Such phenomenon does not occur under reaction at 2 MPa (see Fig S7), it can be speculated that higher pressure gives impetus to the carbon diffusion so the surface dynamic structure may reach new balance during reaction. However, the features of Fe_3O_4 at 660 cm^{-1} that were usually observed in pure Fe_2O_3 catalyst during activation stage were not well resolved in Raman spectra [49,50]. This can be ascribed to the reduced crystallinity and lattice distortion induced by Mn. These findings indicate that CO dissociation occurs during the activation stage, and oxygen is primarily rejected as CO_2, whereas carbon atoms can be hydrogenated into light hydrocarbons or deposited on catalyst surface to postpone CO dissociation. The delayed emergence of carbon species in Fe4Mn1-SCM than Fe4Mn1-CP (350 °C for 3 h vs. 200 °C) indicates a severe carbon deposition and low hydrogenation activity of Fe4Mn1-CP, which is consistent with the results of TPH and XPS.

3.4. Catalytic performance test

Catalytic performance over Fe4Mn1 catalysts were evaluated for at least 24 h time-on-stream. A Mn-free iron catalyst prepared via solution combustion method was also evaluated as a control. As shown in Table 3, a suitable quantity of Mn added to the Fe-SCM showed marginal increase in CO conversion, but suppressed CH_4 formation and enhanced the production of long chain hydrocarbons. Mn generally acts as an electron donor due to its lower work function φ_m, resulting in an increased electron density of neighboring Fe atoms [51]. As demonstrated by the increased O/P ratios in

Table 2
Detailed peak deconvolution data and the surface element analysis results.

Catalysts	δ/(α + β + γ + δ)	δ_2/δ_1	γ_2/γ_1	Surface element distribution[a]			
				Fe	Mn	O	C
Fe4Mn1-SCM	0.69	0.14	1.54	2.59	3.03	15.8	78.6
Fe4Mn1-CP	0.75	0.46	3.15	1.60	1.25	9.25	87.9

[a] The surface element distribution data are obtained from the XPS result.

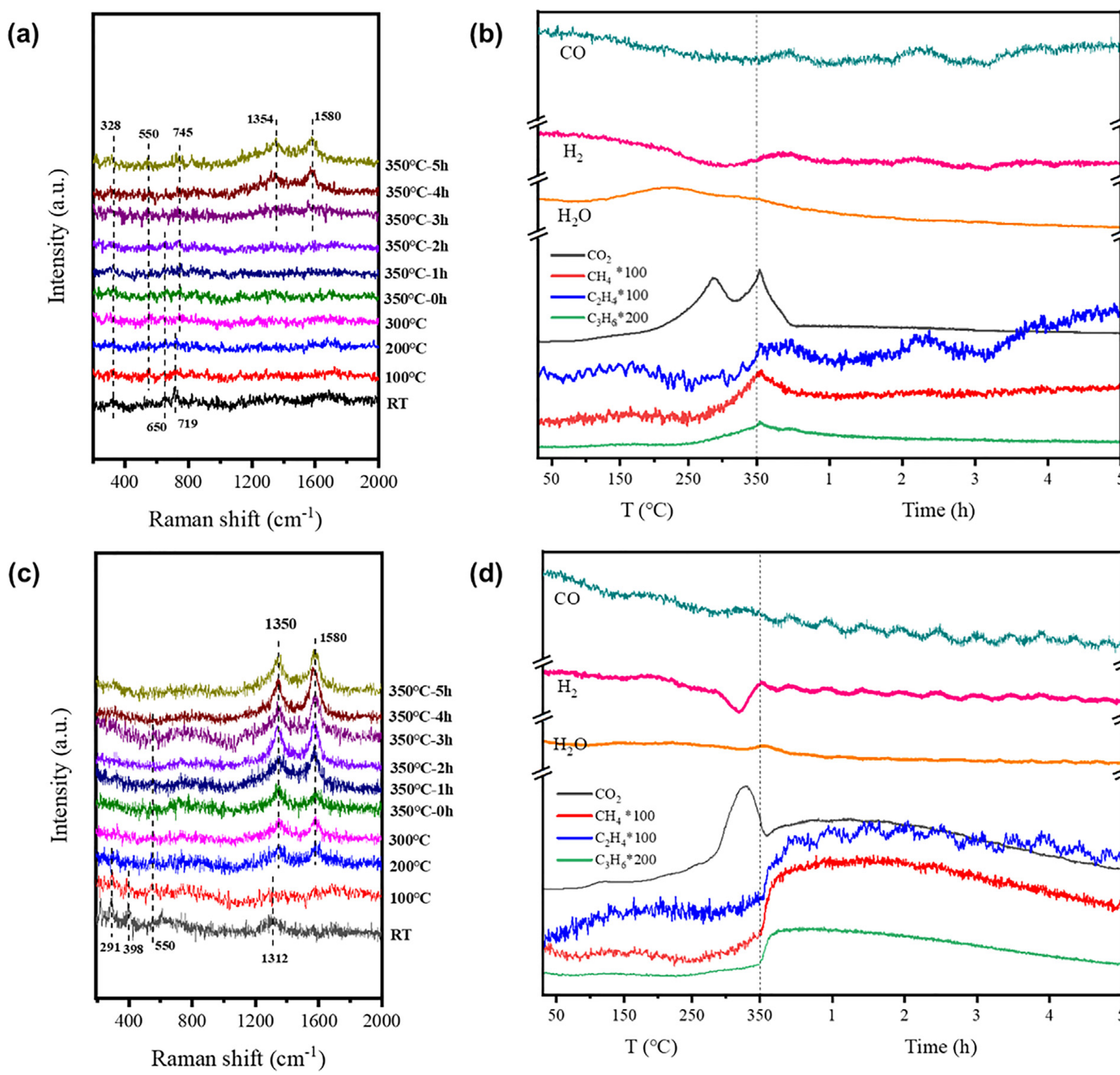


Fig. 11. In-situ Raman-MS profiles of Fe4Mn1 catalysts during activation process of Fe4Mn1-SCM (a, b) and Fe4Mn1-CP (c, d).

Table 3
Catalytic performance of different Fe and Fe4Mn1 catalysts[a].

Catalysts	X_{CO}/%	S_{CO2}/%	S_{HCs} (C-mol%, CO_2 free)				O/P[b]	STY/g·kg_{cat}^{-1}·h^{-1}		Carbon balance/%
			CH_4	C_{2-4}^{0}	$C_{2-4}^{=}$	C_{5+}		$C_{2-4}^{=}$	$C_{2-7}^{=}$	
Fe-SCM	21.0	35.0	21.7	21.2	40.5	16.7	1.91	1038.4	1322.8	97.7
Fe4Mn1-SCM	22.5	24.8	12.9	9.4	32.2	45.5	3.42	920.0	1148.8	97.6
Fe4Mn1-CP	32.4	42.8	19.8	14.0	51.2	15.1	3.65	1505.2	1782.0	98.4

[a] Reaction conditions: 280 °C, 2.0 MPa, 45 %CO/45 %H_2/N_2, 7500 h^{-1}, TOS = 24 h.
[b] The ratio of olefins to paraffins of C_2-C_4.

the Mn-induced catalysts, such electronic modification can facilitate CO dissociative adsorption, C–C coupling reaction, and olefin desorption, hence boosting the yield of olefins. When comparing the Fe4Mn1-SCM and Fe4Mn1-CP catalysts, the Fe4Mn1-CP showed a much higher CO conversion (32.4 %) and higher undesirable C1 by products (CO_2 and CH_4). Moreover, the products obtained from Fe4Mn1-CP were shifted from long-chain hydrocarbons to light hydrocarbons, especially $C_{2\text{-}4}$ olefins. The calculated space time yield (STY) of the light olefins given by Fe4Mn1-CP and Fe4Mn1-SCM was 1505.2 $g{\cdot}kg_{cat}^{-1}{\cdot}h^{-1}$ and 920.0 $g{\cdot}kg_{cat}^{-1}{\cdot}h^{-1}$, respectively, suggesting that the two catalysts had different chain-growth abilities. According to the ICP results, only 0.004 wt% of Na remains in Fe4Mn1-CP, much lower than the amount of Na promoter added in Fe-based catalysts. To exclude the influence of residual Na, a Na-containing catalyst, Fe4Mn1Na-SCM, with a molar ratio of $Fe(NO_3)_3{\cdot}9H_2O/NaNO_3$ = 500/1 was prepared and tested (Table S6). This sample shows similar activity and product selectivity to the Fe4Mn1-SCM, suggesting that the residual Na has no significant impact on the Fe-Mn bimetallic catalysts. We also investigated the FTS performance of Fe4Mn1-PM bimetallic catalyst prepared via physical mixing of Fe_2O_3 and MnO_x nanoparticles synthesized using low-temperature precipitation method. As shown in Table S6, Fe4Mn1-PM shows a higher catalytic activity and a modest O/P ratio of 2.28, revealing a slightly weakened Fe-Mn interaction than Fe4Mn1-CP and Fe4Mn1-SCM.

Detailed hydrocarbon product distribution with different carbon numbers obtained from Fe4Mn1-SCM and Fe4Mn1-CP was illustrated in Fig. 12(a). Interestingly, Fe4Mn1-SCM exhibited a double ASF product distribution with two chain growth probabilities, where α_1 = 0.53 was within C_1-C_9 carbon range and α_2 = 0.79 was for C_{9+} long-chain hydrocarbons. In contrast, hydrocarbon products from Fe4Mn1-CP showed a typical ASF distribution with a much smaller α (0.43). Compared with the Mn-modified Fe-based FTS catalysts reported in literature (Table S5), the prepared Fe4Mn1-SCM catalyst presented higher C_{5+} liquid selectivity while exhibiting lower CO_2 selectivity at moderate CO conversion levels. Catalyst stability tests were performed under the same operating condition and loading quantity as for catalytic performance evaluation over 100 h time-on-stream. As shown in Fig. 12(c, d), Fe4Mn1-CP showed a short induction period followed by a rapid activity loss at 40 h, while Fe4Mn1-SCM exhibited a consistent activity increase and reached to a stable level at 50 h. Such a dramatical change in activity reflects the significant changes occurred in the active centers of the two catalysts.

3.5. Structure-performance relationship

Different $Fe_xC_yO_z$ compositions were fabricated with well-defined Fe-Mn oxides prepared using solution combustion method and co-precipitation method. Under reaction conditions, the composition may further change, resulting in variations in activity and product distribution. To further understand the structure changes of the active centers on product selectivity, kinetic analysis was carried out with Fe4Mn1-SCM and Fe4Mn1-CP catalysts. The results (Fig. S5) demonstrate that the apparent activation energies for CO_2 and CH_4 formation are 150.4 kJ/mol and 97.1 kJ/mol for Fe4Mn1-CP, which are lower than those for Fe4Mn1-SCM (255.8 kJ/mol and 136.3 kJ/mol). This indicates that formation of

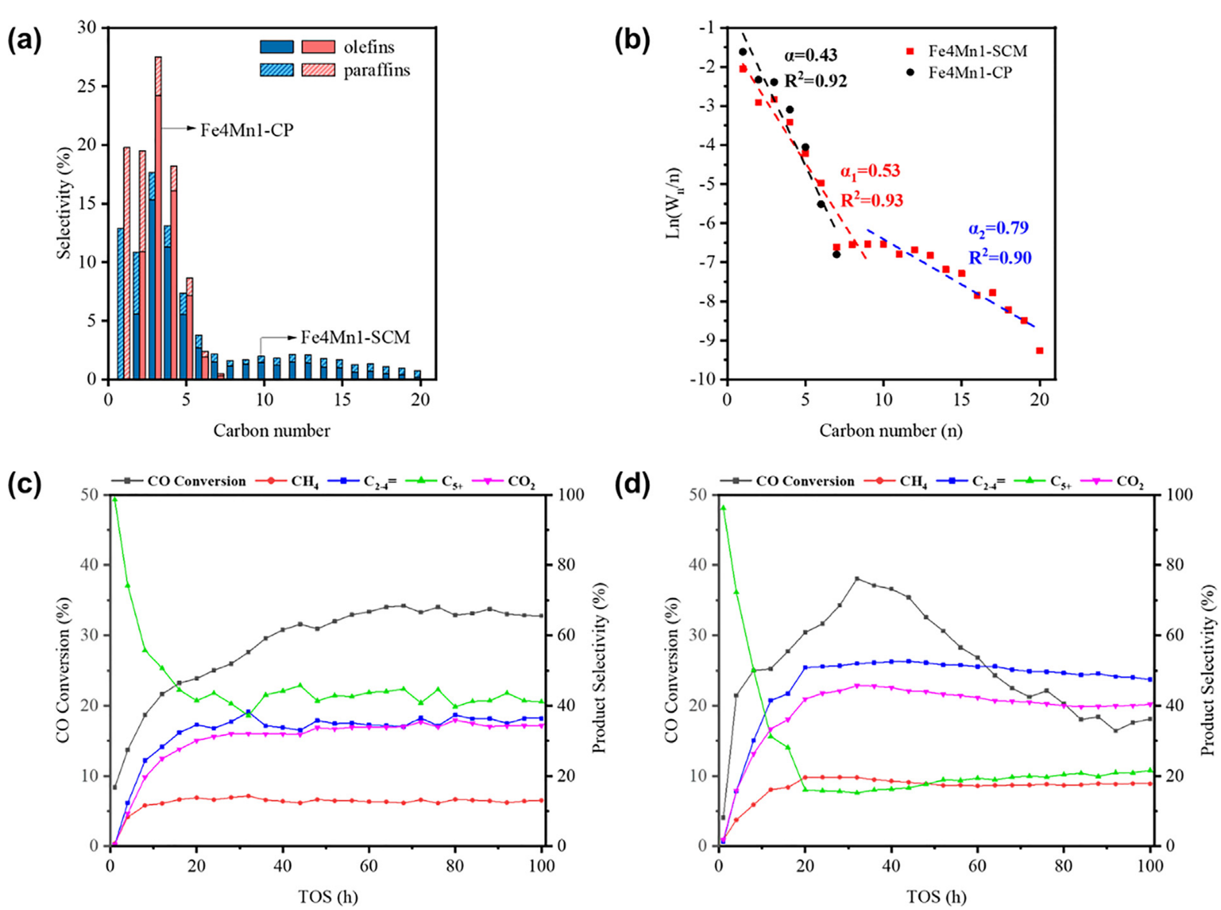


Fig. 12. Carbon number distribution of hydrocarbon products over Fe4Mn1 catalysts (a); Chain growth probability of the Fe4Mn1 catalysts (b); Stability tests of Fe4Mn1-SCM (c) and Fe4Mn1-CP (d) catalysts over 100 h time-on-stream.

C1 by-products is extremely unfavorable on Fe4Mn1-SCM. In addition, the apparent activation energies of C_2 and C_3 hydrocarbons (C_2H_4, C_2H_6, C_3H_6, and C_3H_8) derived from Fe4Mn1-CP are significantly lower than those from Fe4Mn1-SCM, which is consistent with the hydrocarbon distribution.

Surface properties of the spent catalysts were examined using C_3H_6-TPD (Fig. S6). According to the desorption profiles, Fe4Mn1-CP exhibits a substantially smaller portion of strong adsorption feature (210 °C), manifesting that C_3H_6 tends to desorb more easily from the surface of Fe4Mn1-CP. Structure *Operando* Raman/MS was performed (300 °C, 2.0 MPa, $H_2/CO = 1$) to investigate the structure transformation of catalysts during reaction (Fig. S7). Unlike the pure Fe catalyst, no features of Fe_3O_4 except two bands of carbon species (1350 cm^{-1} and 1580 cm^{-1}) are observed on Raman spectra of the Mn-modified Fe-based catalysts [49]. CO conversion and product selectivity of C_1-C_3 hydrocarbons stabilize within 30 min at onset of the reaction. Fe4Mn1-CP shows a higher signal intensity for CO_2 than Fe4Mn1-SCM, which is in line with the observation in catalytic performance evaluation. CH_4 and C_3H_6 are the most two abundant hydrocarbons shown in the product distribution histogram (Fig. S7), with the former representing hydrogenation activity whilst the latter representing the C–C coupling reaction activity. The competition between these two activities is estimated using the ratio of the changes (ε) in the peak intensity of CH_4 and C_3H_6 within the 5 h of the reaction (Eq. (5)).

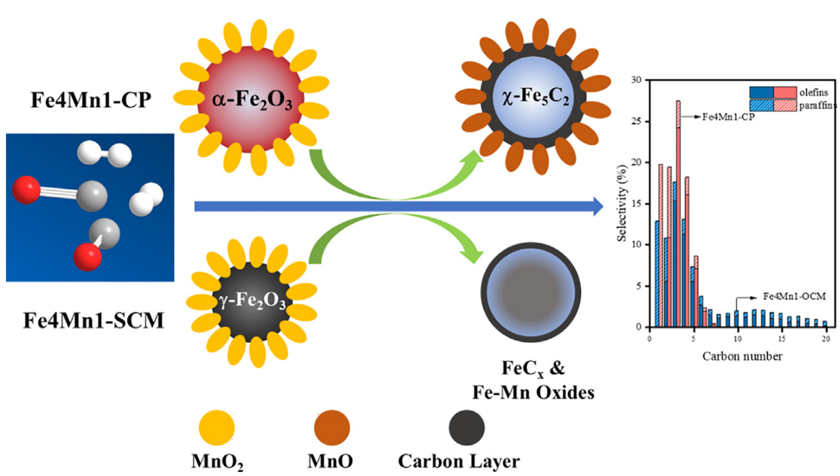
$$\varepsilon = \frac{I_{CH4,5h} - I_{CH4,0h}}{I_{C3H6,5h} - I_{C3H6,0h}} \tag{5}$$

Where $I_{CH4,5h}$, $I_{CH4,0h}$, $I_{C3H6.5h}$, and $I_{C3H6,0h}$ represent the peak intensities of CH_4 and C_3H_6 at TOS = 5 h and 0, respectively. Here, the calculated values of ε for Fe4Mn1-SCM and Fe4Mn1-CP are 2.35 and 2.88, respectively, reflecting that Fe4Mn1-SCM is more active towards C–C coupling, which agrees with the calculated chain growth ability factors (α).

On the basis of the FTS performance evaluation, structural characterization of the spent catalysts, and the *Operando* Raman/MS experiments, a possible reaction mechanism is proposed for olefins with varying carbon distributions from syngas over Fe4Mn1 catalysts (Scheme 1). Mn dopant exerts a strong interplay of structural and electronic effects to iron species. Mn atoms can enter the lattice of the iron oxides, resulting in a more stable Fe-Mn mixed oxide with great resistance to phase transformation under reaction conditions. The electronic effects can be attributed to the increased surface basicity, which results in enhanced CO dissociation and olefin desorption during FTS. In contrast to Fe-Mn catalyst prepared by co-precipitation of Fe and Mn ions, we prepared Fe-Mn catalyst via direct SCM route, which showed higher liquid products yield and lower C1 byproducts on FTS. SCM can be used to fabricate homogenous, and porous, crystalline metal oxide, allowing for more intimate Fe-Mn interaction and hence the development of more stable Fe-Mn oxides. During activation process, the resultant Fe-Mn mixed oxide precursor may only be partially carburized with a combination of Fe-Mn mixed oxides ($Fe_xMn_{3-x}O_4$, $Fe_xMn_{1-x}O_x$) and carbides (χ-Fe_5C_2, ε-$Fe_{2.2}C$). The ε-$Fe_{2.2}C$ phase is well recognized for its superior selectivity to C_{5+} hydrocarbons and ability to inhibit CH_4 formation. CO_2 formation on Fe-based FTS catalysts consists of primary CO_2 ($2CO \rightarrow C + CO_2$) and secondary CO_2 (WGS, $CO + H_2O \rightarrow CO_2 + H_2$). According to Hensen et al [7], ε-$Fe_{2.2}C$ was not active for production of primary CO_2 within temperature range of 220–265 °C. This can explain why the Fe4Mn1-SCM catalyst has a lower CO_2 selectivity. In contrast, the *Operando* Raman spectra and Mossbauer results show that Fe4Mn1-CP has a faster carburization rate and higher carburization degree than Fe4Mn1-SCM. Moreover, no Fe-Mn mixed oxide was detected in the spent Fe4Mn1-CP, indicating that Mn atoms were not incorporated into the lattice of the iron oxides during activation. The oxygen vacancies in partially reduced MnO_x can provide adsorption sites for O atoms in CO, therefore activating the C = O bond and enhancing CO dissociation [52]. The C atoms from the dissociated CO molecules can migrate onto Fe surface for carburization. Thus, it is hypothesized that Mn promoter mostly exists in the form of partially reduced MnO_x, which enhances CO dissociation and surface basicity, thus altering the product distribution to short chain hydrocarbons. As a result, the interaction between the active Fe metal and Mn promoters can be finely tuned to modify the product distribution in FTS.

4. Conclusion

The FTO activity and product selectivity of Mn-promoted bulk Fe catalysts prepared by a facile solution combustion method (SCM) and a co-precipitation method (CP) were investigated. SCM can be utilized to create a highly dispersed and loose metal oxide nanoparticles with abundant interfacial Fe-MnO_x structure. The Fe-Mn catalyst prepared via SCM showed a C_{5+} selectivity of 45.5 % under a CO conversion of 22.5 %. Unlike Fe4Mn1-SCM, the Fe4Mn1-CP catalyst mainly produced short chain hydrocarbons (selectivity of C_1-C_4 is 85 %). The difference in catalytic performance, especially the hydrocarbon distribution, might be attributed to the distinct promoting effects of Mn on iron active centers; that is, Mn acts more like a structural promoter in



Scheme 1. Proposed reaction mechanism for olefins production from syngas over Fe4Mn1 catalysts with different proximity between Fe (red spheres) and Mn (blue spheres).

Fe4Mn1-SCM whilst exerting largely electronic effects in Fe4Mn1-CP. In more detail, the as-prepared Fe-Mn catalyst showed a lower content of Fe_5C_2 in activation and reaction process due to the stabilization of Fe-Mn mixed oxide. The intimate contact between Fe and Mn slows down the carbon diffusing but allows for sufficient time for lattice expansion, resulting in a kinetically favorable in forming carbon-rich carbide, i.e., the ε-$Fe_{2.2}C$ phase. In terms of surface chemistry, strong Fe-Mn interactions also retard the carbon deposition and WGS reaction. As a result, Fe4Mn1-SCM exhibits excellent catalytic stability over 100 h time-on-stream and more importantly, the CO_2 selectivity is only 24.8 %, which is almost 1/3 less than that of Fe4Mn1-CP. These demonstrated that the interplay of structural and electronic effects induced by Mn can be carefully adjusted via finely controlling the interface structure, hence tailoring the product distribution of FTS.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors greatly acknowledge the funding support from Shanghai Sailing Program (19YF1411000) and Open Fund of State Key Laboratory of Chemical Engineering (GZA01220102).

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jcat.2022.12.003.

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Supplementary Materials

Revisiting the Syngas Conversion to Olefins over Fe-Mn Bimetallic System: Insights from the Promxity Effects

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DFT computational models and methods

The calculated Fe_3O_4 is a cubic crystal with an inverse spinel structure and the FeO is a cubic crystal adopting a rocksalt structure. Their optimized unit cells show a lattice constant of 8.381 Å and 4.308 Å, respectively, which agree well with the experimental value. Surface reactions were calculated on a p (1 × 2) supercell of Fe_3O_4 (311) surface and a p (2 × 3) supercell of FeO (200) surface. To match the experimental $Mn_{0.43}Fe_{2.57}O_4$, 4 Fe atoms in Fe_3O_4 supercell were replaced by Mn. In order to account for the lattice expansion caused by Mn, a FeO single cell with two Fe atoms replaced by Mn was individually optimized, and the lattice constant is 4.401 Å. The supercell model, similar to that of FeO (200) slab, was developed and is used to simulate the experimental $Fe_{0.497}Mn_{0.503}O$ phase. To eliminate the interaction between the repeating slabs, the vacuum gap spacing for all models were set to 15 Å. The calculated pure or Mn-substituted Fe_3O_4 (311) model has 24 Fe or Mn atoms and 32 O atoms, with 12 Fe and 16 O atoms at the bottom fixed, whereas the pure or Mn-substituted FeO (200) model has 36 Fe or Mn atoms and 36 O atoms, with the bottom 24 atoms fixed. The top and side views of the calculated slab models are shown in Fig S3.

All calculations were performed by using the Vienna Ab initio Simulation Package (VASP) with the projected augmented wave (PAW) pseudopotentials. The electron exchange and correlation energies were treated with the Perdew-Burke-Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) with a kinetic energy cutoff of 400 eV [1]. DFT + U using the simplified rotationally invariant approach introduced by Dudarev et al. [2], with U_{eff} = 3.8 eV for Fe and U_{eff} = 4.5 eV for Mn were chosen to describe the 3d states of Fe and Mn [3, 4]. A Gaussian smearing with sigma = 0.2 eV was employed to converge the electronic self-consistent cycles. K-point grids of 4×4×4 and 7×7×7 for the Fe_3O_4 and FeO single crystals, and 3 × 2 × 1 for the slab models were used to sample the Brillouin zones. The force criterion for geometry optimization was set to be < 0. 05 eV/Å.

The formation energy of a surface oxygen vacancy by reacting with H_2 is calculated via eq 1:

$$\Delta E_{ov} = E_{slab-ov} + E_{H_2O} - \left(E_{slab} + E_{H_2}\right) \tag{1}$$

Where E_{slab} and $E_{slab-ov}$ are the total energies of the slabs without and with an oxygen vacancy, E_{H_2} and E_{H_2O} are the total energies of the isolated H_2 and H_2O molecules.

The penetration energy of a C atom from CO with O reacting with H_2 is defined in eq 2:

$$\Delta E_C = E_{slab-C} + E_{H_2O} - \left(E_{slab-ov} + E_{H_2} + E_{CO}\right) \tag{2}$$

Where E_{slab-C} is the total energy of the slab with carbon atom embedded into the oxygen vacancy, E_{CO} is the total energy of the isolated CO molecule.

As such, the carburization of Fe oxides can be seen as the combination of the above two processes, so that the total reaction energy is calculated as the sum of ΔE_{ov}

and ΔE_C:

$$\Delta E_{total} = \Delta E_{ov} + \Delta E_C = E_{slab-C} + 2E_{H_2O} - \left(E_{slab} + 2E_{H_2} + E_{CO}\right) \qquad (3)$$

Table S1. Residual contents of C and H in Fe4Mn1-SCM catalysts with different combustion time.

Time/h	C/%	H/%
0.5	0.315	0.478
2	0.11	0.346
4	0.11	0.344

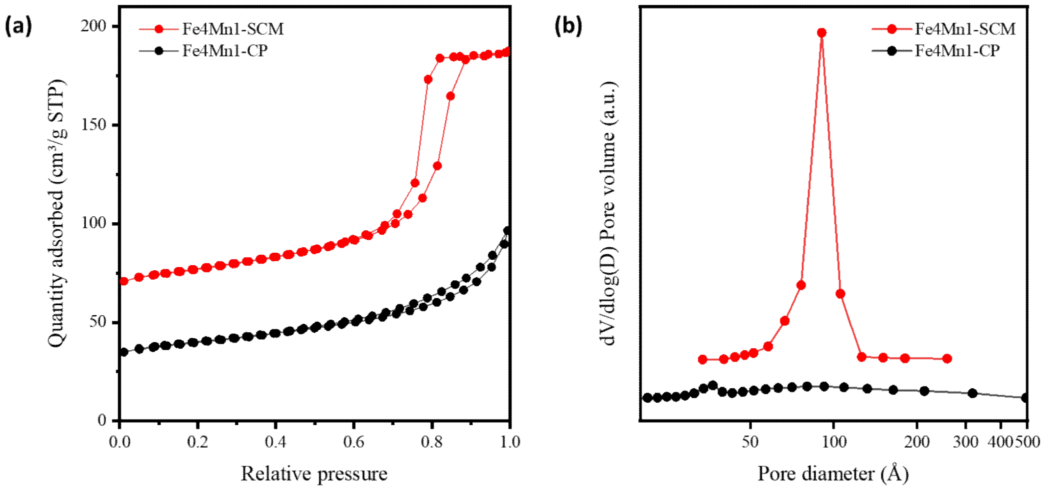

Fig S1. (a) Adsorption-desorption isotherms of different Fe4Mn1 catalysts. (b) Pore size distribution of different Fe4Mn1 catalysts.

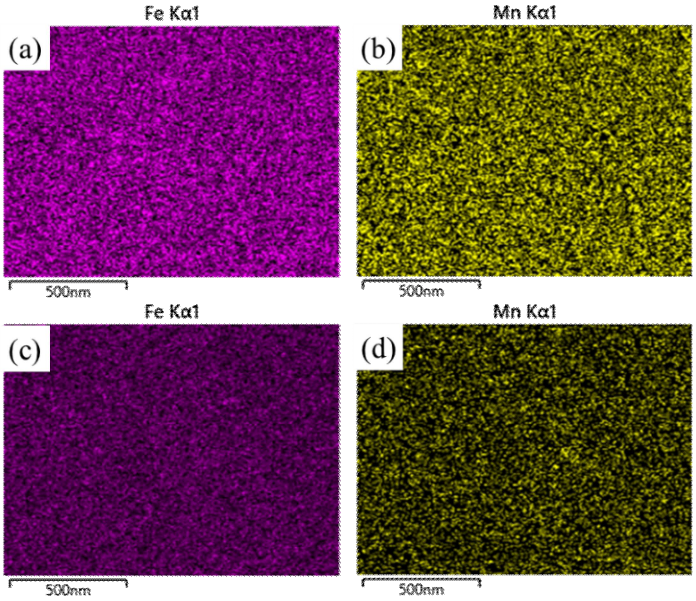

Fig S2. Elemental mapping of Fe4Mn1-SCM (a, b) and Fe4Mn1-CP (c, d).

Table S2. Detailed MES results of activated Fe4Mn1 catalysts.

	Hhf / KOe	IS / $mm \cdot s^{-1}$	QS / $mm \cdot s^{-1}$	Area	Phases
Fe4Mn1-CP	—	0.21	0.95	0.021	Fe^{3+}
	—	1.11	0.96	0.056	Fe^{2+}
	210.88	0.26	-0.08	0.36	
	184.55	0.22	0	0.432	Fe_5C_2
	110	0.2	-0.07	0.132	
Fe4Mn1-SCM	—	0.41	0.8	0.141	Fe^{3+}
	—	0.97	0.78	0.183	Fe^{2+}
	215.89	0.24	-0.08	0.251	
	183.23	0.22	-0.04	0.238	Fe_5C_2
	102.96	0.23	0.37	0.138	
	201.17	0.22	-0.04	0.09	Fe_3C

Table S3. Detailed FFT analysis results of spent Fe4Mn1 catalysts.

	Phase		R1 / Å	R2 / Å	R3 / Å	<R1,R2>	<R1,R3>	<R2,R3>
Fe4Mn1-SCM	Fe_5C_2 89-8968	FFT	2.074	1.679	1.771	117.858°	68.837°	49.021°
		PDF	2.083	1.643	1.773	118.974°	70.821°	48.153°
	$Mn_{0.43}Fe_{2.57}O_4$ 89-2807	FFT	2.985	3.032	2.125	89.885°	44.496°	45.389°
		PDF	2.988	2.988	2.113	90°	45°	45°
Fe4Mn1-CP	Fe_5C_2 89-8968	FFT	2.025	4.261	1.977	100.713°	27.124°	73.589°
		PDF	2.028	4.253	2.012	102.787°	27.471°	75.316°
	Mn_5O_8 72-1427	FFT	3.636	1.985	2.022	107.829°	75.864°	31.965°
		PDF	3.679	1.954	2.035	109.768°	78.407°	31.361°

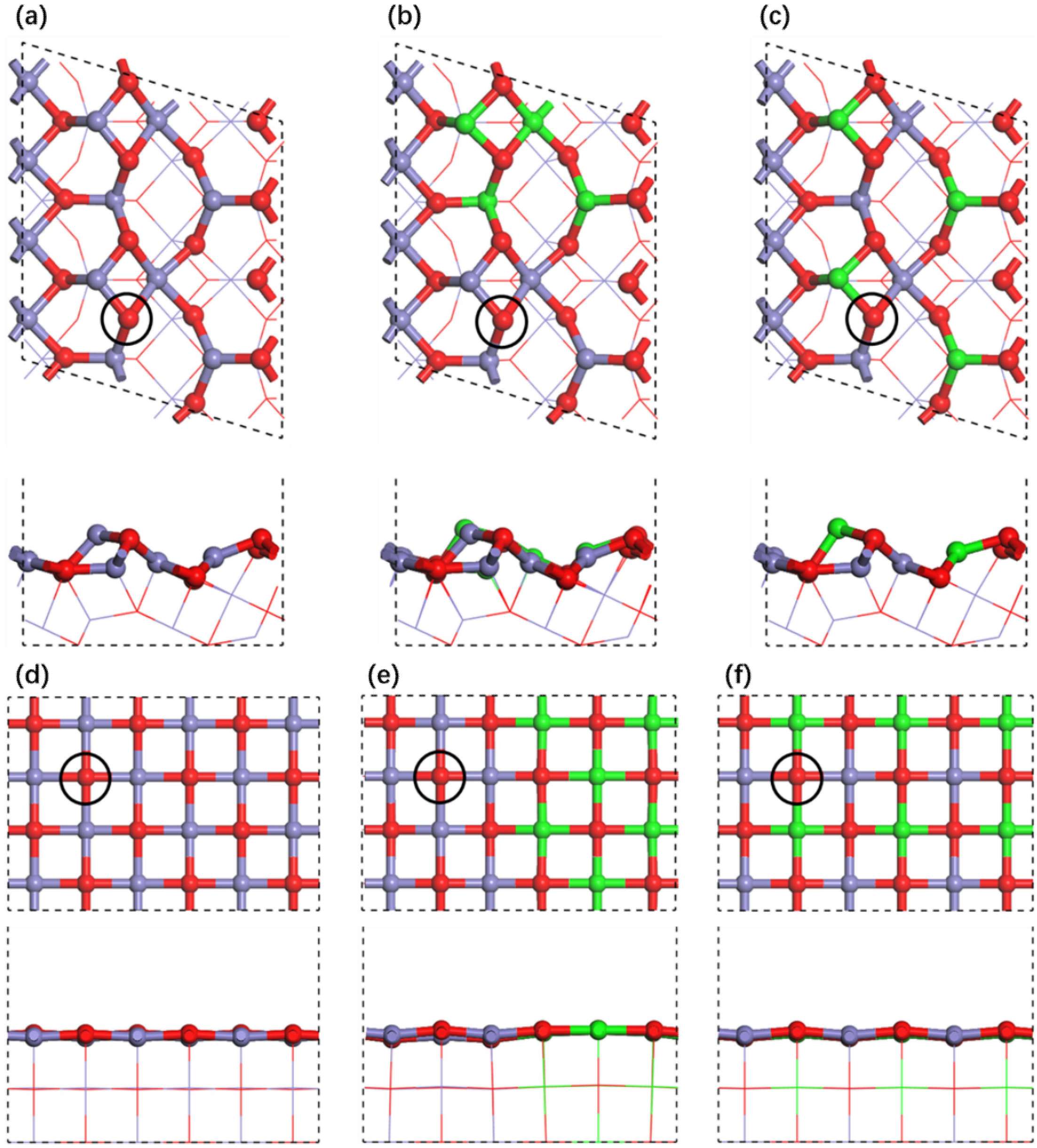

Fig S3. Top and side views of (a) Fe_3O_4, (b) Fe_3O_4_Mn_3O_4, (c) $(Fe,Mn)_3O_4$ and (d) FeO, (e) FeO_MnO, (f) (Fe,Mn)O (Blue: Fe atoms; red: O atoms; green: Mn atoms). O atoms in the black circles are removed and substituted by C atoms.

Table S4. Detailed MES results of spent Fe4Mn1 catalysts.

	Hhf / KOe	IS / $mm\cdot s^{-1}$	QS / $mm\cdot s^{-1}$	Area	Phases
Fe4Mn1-CP	—	0.39	1.07	0.173	Fe^{3+}
	216.62	0.26	-0.12	0.302	Fe_5C_2
	185.68	0.2	0	0.349	
	112.72	0.25	0.06	0.176	
Fe4Mn1-SCM	—	0.42	0.94	0.145	Fe^{3+}
	—	1.08	0.67	0.06	Fe^{2+}
	494.75	0.32	-0.01	0.07	Fe_3O_4
	460.83	0.54	0.07	0.153	
	218.93	0.24	-0.11	0.208	Fe_5C_2
	192.78	0.2	0.01	0.205	
	113.69	0.19	-0.09	0.107	
	172.99	0.19	-0.13	0.051	$Fe_{2.2}C$

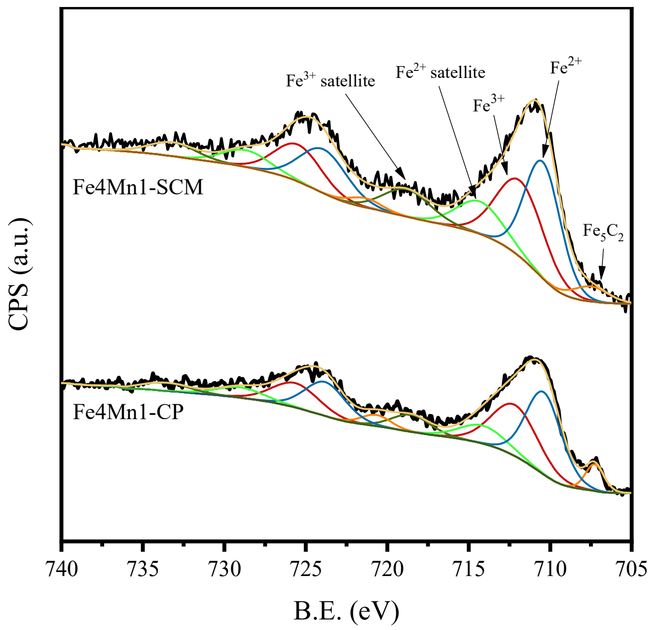

Fig S4. XPS spectra of spent Fe4Mn1 catalysts.

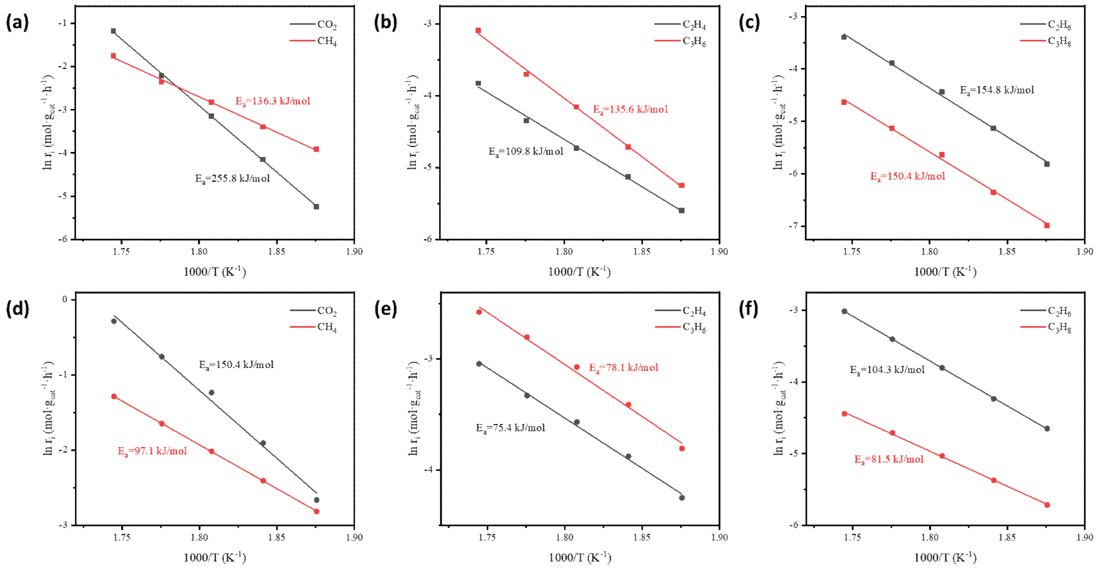

Fig S5. Activation energy (E_a) of C_1-C_3 products for Fe4Mn1-SCM (a-c) and Fe4Mn1-CP (d-f). Kinetic measurements were taken at the following conditions: 260-300 °C, 2.0 MPa, 50ml/min, 15mg catalyst diluted with 85mg SiC

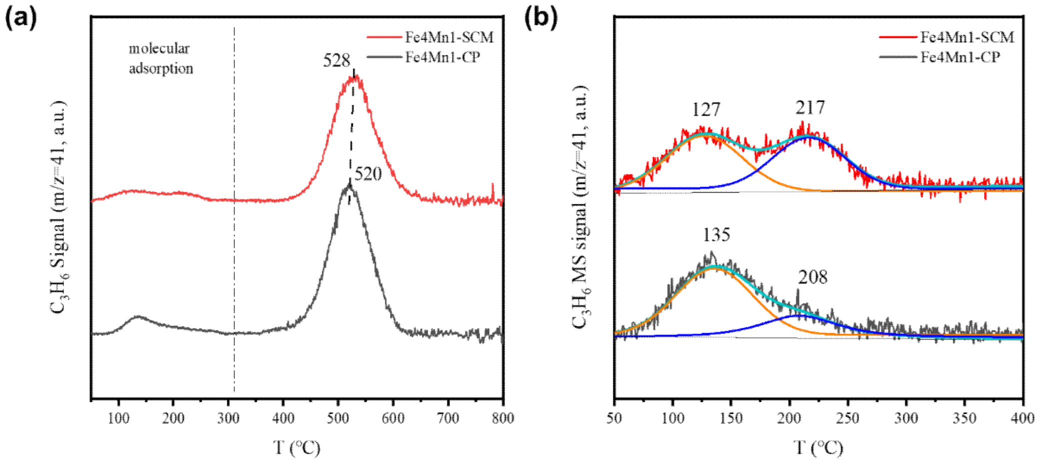

Fig S6. C_3H_6-TPD profiles of the reduced Fe4Mn1 catalysts.

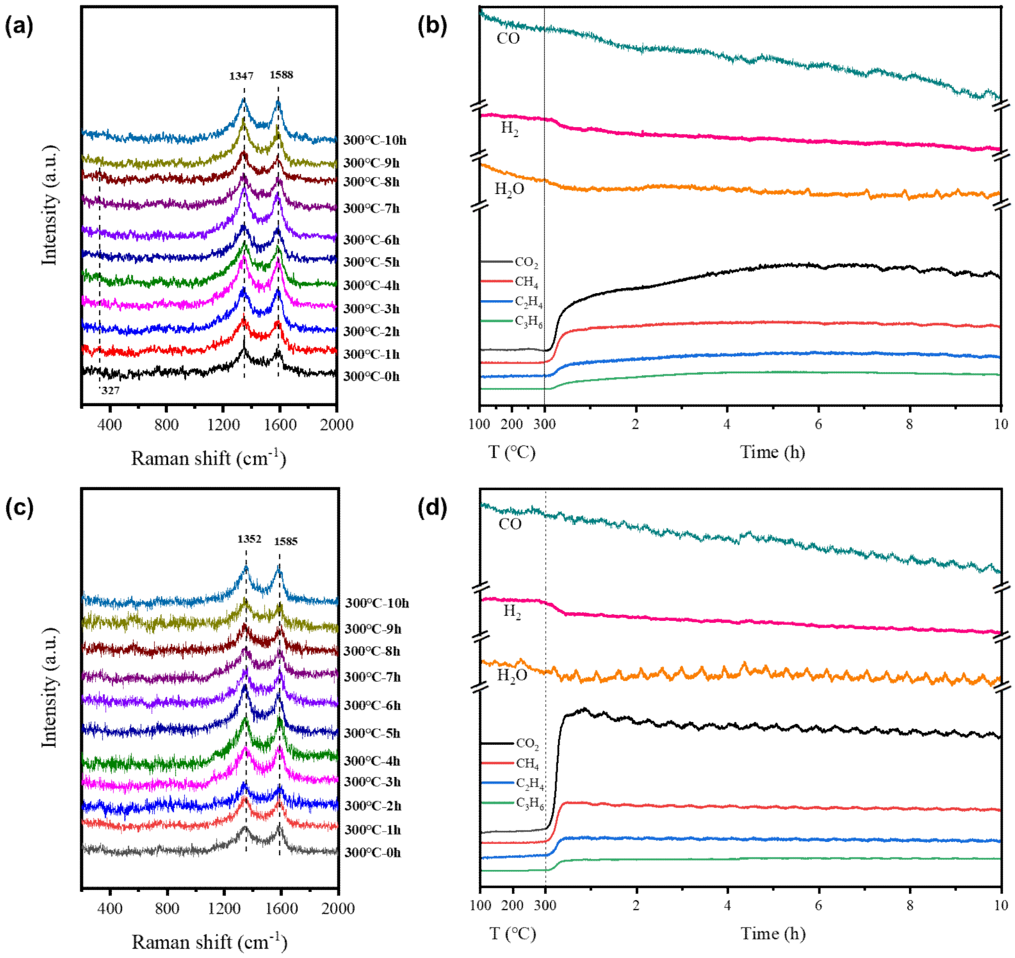

Fig S7. In-situ Raman-MS profiles of Fe4Mn1 catalysts during reaction process of Fe4Mn1-SCM (a, b) and Fe4Mn1-CP (c, d).

Table S5. Comparison of catalytic performance with the previous works.

Cat.	X_{CO} (%)	S_{CO2} (%)	S_{HCs} (%) $C_{2-4}^{=}$	C_{2-4}^{0}	C_{5+}	olefins	FTY ($\mu mol_{CO}\cdot g_{Fe}^{-1}\cdot s^{-1}$)	Ref.
MnO_2/α-Fe_2O_3	48.2	29.4	51.1	8.9	26.6	-	181	[5]
Al-Fiber@meso-Al_2O_3@Fe-Mn-K	65.6	33.1	32.3	8.1	40.8	-	149	[6]
Fe-10MnK-AC	85	48	39.4	8.13	29.7	-	150	[7]
Fe/CeO_2@0.2MnO_2	39.9	40.3	40.4	7.2	33.6	-	119	[8]
FeMn@SiO_2-0.25	67.2	47.5	20.9		71.8	-	-	[9]
FeMn/BN	19.8	7.5	43.4		37.0	-	7.86	[10]
FeMn10	83	34	26.6		63.9	-	23.2	[11]
0.5Mn/γ-Fe_2O_3	63-68	36.8	58.4	9.5	21.2	-	23.1	[12]
FeMn-3.0Cu	96.9	23	40.1	30.4	9.5	-	5.39	[13]
6wt%Mn/Fe_3O_4	41.5	37.8	60.1	6.5	23.7	-	16.8	[14]
FeMn@Si-c	56.1	13	-	-	-	75	16.8	[15]
Fe4Mn1-SCM	22.5	24.8	32.2	9.4	45.5	54.4	132	TW
Fe4Mn1-CP	32.	46.	51.	14.	15.	60.6	189	T

4	8	2	0	1	W

"TW" is short for "this work".

Table S6. Supplementary catalytic performance of different Fe and Fe4Mn1 catalysts[a]

Catalysts	X_{CO}/%	S_{CO2}/ %	S_{HCs} (C-mol%, CO_2 free) CH_4	C_{2-4}^{0}	$C_{2-4}^{=}$	C_{5+}	O/P[b]	STY/g·kg_{cat}^{-1}·h^{-1} $C_{2-4}^{=}$	$C_{2-7}^{=}$	Carbon balance/%
Fe-SCM	21.0	35.0	21.7	21.2	40.5	16.7	1.91	1038.4	1322.8	97.7
Fe4Mn1-SCM	22.5	24.8	12.9	9.4	32.2	45.5	3.42	920.0	1148.8	97.6
Fe4Mn1-CP	32.4	42.8	19.8	14.0	51.2	15.1	3.65	1505.2	1782.0	98.4
Fe4Mn1-PM	34.9	44.8	22.2	18.4	42.1	17.3	2.28	1371.2	1748.5	97.1
Fe4Mn1Na-SCM	26.3	30.7	12.0	8.2	31.2	48.6	3.79	961.3	1173.2	97.0

[a] Reaction conditions: 280 °C, 2.0 MPa, 45%CO/45%H_2/N_2, 7500 h^{-1}, TOS=24 h.

[b] The ratio of olefins to paraffins of C_2-C_4.

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